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Cell – The Unit of Life

NEET Biology · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	193 questions
Answer key	at the end

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MCQ (Single Correct Answer) 193 questions

Q1. RE-NEET 2026

Mitochondrial inner membrane encloses ____ .

(A) aqueous humor (B) matrix (C) cytosol (D) mucus

Open solution

Q2. RE-NEET 2026

Cell theory was formulated by ____

(A) Antonie Von Leeuwenhoek (B) Schleiden and Schwann

(C) Robert Brown (D) Singer and Nicolson

Open solution

Q3. RE-NEET 2026

The plastid that stores xanthophyll is known as ____ .

(A) amyloplast (B) chloroplast (C) chromoplast (D) aleuroplast

Open solution

Q4. RE-NEET 2026

Smooth endoplasmic reticulum ____ .

(A) is a site for the synthesis of carbohydrates

(B) has ribosomes attached to its surface

(C) is the major site for the synthesis of lipids

(D) is actively involved in protein synthesis

Open solution

Q5. RE-NEET 2026

Which of the following are characteristics of prokaryotic cells?

(a) Ribosomes are made of 50 S and 30 S subunits

(b) They can have plasmids

(c) They contain mesosome

(d) They have peroxisomes

Choose the correct answer from the options given below :

(A) (a), (b) and (c) only (B) (b) and (c) only

(C) (a) and (c) only (D) (a), (c) and (d) only

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Q6. RE-NEET 2026

Match List-I with List-II.

	List-I		List-II
A.	Cristae	I.	Flat membrane sacs in stroma of chloroplast
B.	Cisternae	II.	Infoldings in mitochondria
C.	Thylakoids	III.	Cell membrane
D.	Phospholipid	IV.	Disc shaped sacs in the Golgi apparatus

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II
(B) A-III, B-IV, C-I, D-II
(C) A-II, B-IV, C-I, D-III
(D) A-II, B-IV, C-III, D-I

Open solution

Q7. RE-NEET 2026

Endomembrane system includes ____ .

(A) Golgi complex, chloroplast, peroxisomes and vacuole
(B) endoplasmic reticulum, Golgi complex, lysosomes and vacuole
(C) endoplasmic reticulum, chloroplast, peroxisomes and vacuole
(D) mitochondria, chloroplast, peroxisomes and vacuole

Open solution

Q8. NEET 2026

Which one of the following is the site for active ribosomal RNA synthesis?

(A) Centrosome (B) Chromatin (C) Nucleolus (D) Kinetochore

Open solution

Q9. NEET 2026

Non-membrane bound cell organelles found in both prokaryotic and eukaryotic cells are ____ .

(A) Ribosomes (B) Lysosomes (C) Centrosomes (D) Mitochondria

Open solution

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Q10. NEET 2026

Open solution

Choose the correct statements regarding cell organelles and their inclusions.

A. The endomembrane system includes Golgi complex, endoplasmic reticulum and mitochondria.

B. Rough endoplasmic reticulum bears ribosomes on its surface.

C. Both mitochondria and plastids have circular DNA.

D. A network of microtubules, microfilaments and intermediate filaments present in the cytoplasm is called cytoskeleton.

E. Mitochondrion is a single membrane-bound structure.

Choose the correct answer from the options given below :

(A) A and B only (B) A, B and C only (C) C, D and E only (D) B, C and D only

Q11. NEET 2026

Open solution

Select the correct statements regarding cell membrane in eukaryotic cell.

A. Membrane of human RBCs has approximately 52% protein.

B. Major phospholipids are arranged in a bilayer.

C. Extensions of the plasma membrane into the cell form mesosomes.

D. Tails towards the inner part of lipids are hydrophobic and thus protected from aqueous medium.

E. Glycocalyx is present on the outer surface of the plasma membrane.

Choose the correct answer from the options given below:

(A) C, D and E only (B) B, C and E only (C) A, B and D only (D) A, C and E only

Q12. NEET 2025

Open solution

Given below are two statements : one is labelled as Assertion (A), and the other is labelled as Reason (R).

Assertion (A) : The primary function of the Golgi apparatus is to package the materials made by the endoplasmic reticulum and deliver it to intracellular targets and outside the cell.

Reason (R) : Vesicles containing materials made by the endoplasmic reticulum fuse with the cis face of the Golgi apparatus, and they are modified and released from the trans face of the Golgi apparatus.

In the light of the above statements, choose the correct answer from the options given below :

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

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Q13. NEET 2025

From the statements given below choose the correct option :

A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.

B. Each ribosome has two sub-units.

C. The two sub-units of 80 S ribosome are 60 S and 40 S while that of 70 S are 50 S and 30 S .

D. The two sub-units of 80 S ribosome are 60 S and 20 S and that of 70 S are 50 S and 20 S .

E. The two sub-units of 80 S are 60 S and 30 S and that of 70 S are 50 S and 30 S .

(A) A, B, E are true **(B)** B, D, E are true **(C)** A, B, C are true **(D)** A, B, D are true

Open solution

Q14. NEET 2025

A specialised membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is

(A) Cristae **(B)** Endoplasmic Reticulum

(C) Mesosome **(D)** Chromatophores

Open solution

Q15. NEET 2025

Match List-I with List- II.

	List - I		List - II
A	Centromere	I	Mitochondrion
B	Cilium	II	Cell division
C	Cristae	III	Cell movement
D	Cell membrane	IV	Phospholipid Bilayer

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-III, D-I **(B)** A-II, B-III, C-I, D-IV

(C) A-I, B-II, C-III, D-IV **(D)** A-II, B-I, C-IV, D-III

Open solution

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Q16. NEET 2024 (Re-Examination)

Open solution

Match List I with List II :

	List - I		List - II
(A)	Flemming	(I)	Disc shaped sacs or cisternae near cell nucleus
(B)	Robert Brown	(II)	Chromatin
(C)	George Palade	(III)	Ribosomes
(D)	Camillo Golgi	(IV)	Nucleus

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-III, D-I

(B) A-II, B-III, C-I, D-IV

(C) A-I, B-II, C-III, D-IV

(D) A-IV, B-II, C-III, D-I

Q17. NEET 2024 (Re-Examination)

Open solution

Mesosome in a cell is a :

(A) Membrane bound vesicular structure

(B) Chain of many ribosomes attached to a single mRNA

(C) Special structure formed by extension of plasma membrane

(D) Medium sized chromosome

Q18. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II:

	List - I		List - II
(A)	Histones	(I)	Loosely packed chromatin
(B)	Nucleosome	(II)	Densely packed chromatin
(C)	Euchromatin	(III)	Positively charged basic proteins
(D)	Heterochromatin	(IV)	DNA wrapped around histone octamer

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-III, B-I, C-IV, D-II

(C) A-II, B-III, C-IV, D-I

(D) A-III, B-IV, C-I, D-II

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Q19. NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	F_1 Particles	(I)	Chromosomes
(B)	Histones	(II)	Cillia
(C)	Axoneme	(III)	Golgi apparatus
(D)	Cisternae	(IV)	Mitochondria

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III

(B) A-IV, B-I, C-II, D-III

(C) A-IV, B-I, C-III, D-II

(D) A-IV, B-III, C-I, D-II

Q20. NEET 2024 (Re-Examination)

Open solution

Given below are two statements :

Statement I : Concentrically arranged cisternae of Golgi complex are arranged near the nucleus with distinct convex cis or maturing and concave trans or forming face.

Statement II : A number of proteins are modified in the cisternae of Golgi complex before they are released from cis face.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Q21. NEET 2024

Open solution

Match List I with List II

	List - I		List - II
A.	Nucleolus	I.	Site of formation of glycolipid
B.	Centriole	II.	Organization like the cartwheel
C.	Leucoplasts	III.	Site of active ribosomal RNA synthesis
D.	Golgi apparatus	IV.	For storing nutrients

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-II, B-III, C-I, D-IV

(C) A-III, B-IV, C-II, D-I

(D) A-I, B-II, C-III, D-IV

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Q22. NEET 2024

The DNA present in chloroplast is:

(A) Linear, double stranded
(B) Circular, double stranded
(C) Linear, single stranded
(D) Circular, single stranded

Open solution

Q23. NEET 2024

Match List I with List II :

	List - I		List - II
A.	Axoneme	I.	Centriole
B.	Cartwheel pattern	II.	Cilia and flagella
C.	Crista	III.	Chromosome
D.	Satellite	IV.	Mitochondria

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-II, D-I
(B) A-IV, B-II, C-III, D-I
(C) A-II, B-IV, C-I, D-III
(D) A-II, B-I, C-IV, D-III

Open solution

Q24. NEET 2024

Given below are two statements:

Statement I: Mitochondria and chloroplasts are both double membrane bound organelles.

Statement II: Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct.
(B) Both Statement I and Statement II are incorrect.
(C) Statement I is correct but Statement II is incorrect.
(D) Statement I is incorrect but Statement II is correct.

Open solution

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Q25. NEET 2023 Manipur

Open solution

Given below are two statements :

Statement I :- In bacteria, the mesosomes are formed by the extensions of plasma membrane.

Statement II :- The mesosomes, in bacteria, help in DNA replication and cell wall formation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is correct but Statement II is incorrect.

(B) Statement I is incorrect but Statement II is correct.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

Q26. NEET 2023 Manipur

Open solution

Which of the following statements are correct with respect to Golgi apparatus ?

(A) It is the important site of formation of glycoprotein and glycolipids.

(B) It produces cellular energy in the form of ATP.

(C) It modifies the protein synthesized by ribosomes on ER.

(D) It facilitates the transport of ions.

(E) It provides mechanical support.

Choose the most appropriate answer from the options given below :

(A) (B) and (C) only **(B)** (A) and (C) only **(C)** (A) and (D) only **(D)** (D) and (E) only

Q27. NEET 2023

Open solution

How many different proteins does the ribosome consist of?

(A) 60 **(B)** 40 **(C)** 20 **(D)** 80

Q28. NEET 2023

Open solution

Which of the following are NOT considered as the part of endomembrane system?

A. Mitochondria

B. Endoplasmic reticulum

C. Chloroplasts

D. Golgi complex

E. Peroxisomes

Choose the most appropriate answer from the options given below:

(A) A, C and E only **(B)** A and D only **(C)** A, D and E only **(D)** B and D only

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Q29. NEET 2023

Which of the following functions is carried out by cytoskeleton in a cell?

(A) Protein synthesis
(B) Motility
(C) Transportation
(D) Nuclear division

Open solution

Q30. NEET 2022 Phase 2

Match List-I with List-II

	List - I		List - II
(a)	Porins	(i)	Pink coloured nodules
(b)	leg haemoglobin	(ii)	Lumen of thylakoid
(c)	H^+ accumulation	(iii)	Amphibolic pathway
(d)	Respiration	(iv)	Huge pores in outer membrane of mitochondria

Choose the correct answer from the options given below.

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(B) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(C) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(D) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

Open solution

Q31. NEET 2022 Phase 2

Which type of substance would face difficulty to pass through the cell membrane?

(A) Substance soluble in lipids
(B) Substance with hydrophobic moiety
(C) Substance with hydrophilic moiety
(D) All substances irrespective of hydrophobic and hydrophilic moiety

Open solution

Q32. NEET 2022 Phase 2

If the pH in lysosomes is increased to alkaline, what will be the outcome?

(A) Lysosomal enzymes will be more active
(B) Hydrolytic enzymes will function more efficiently
(C) Hydrolytic enzymes will become inactive
(D) Lysosomal enzymes will be released into the cytoplasm

Open solution

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Q33. NEET 2022 Phase 1

Open solution

Match List I with List II.

	List - I		List - II
(a)	Metacentric chromosome	(i)	Centromere situated close to the end forming one extremely short and one very long arms
(b)	Acrocentric chromosome	(ii)	Centromere at the terminal end
(c)	Submetacentric	(iii)	Centromere in the middle forming two equal arms of chromosomes
(d)	Telocentric chromosome	(iv)	Centromere slightly away from the middle forming one shorter arm and one longer arm

Choose the correct answer from the options given below:

(A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
(B) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
(D) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Q34. NEET 2022 Phase 1

Open solution

Which of the following statements with respect to Endoplasmic Reticulum is incorrect?

(A) RER has ribosomes attached to ER
(B) SER is devoid of ribosomes
(C) Only RER are present in prokaryotes
(D) SER are the sites for lipid synthesis

Q35. NEET 2021

Open solution

Which of the following is an incorrect statement?

(A) Nuclear pores act as passages for proteins and RNA molecules in both directions between nucleus and cytoplasm
(B) Mature sieve tube elements possess a conspicuous nucleus and usual cytoplasmic organelles
(C) Microbodies are present both in plant and animal cells
(D) The perinuclear space forms a barrier between the materials present inside the nucleus and that of the cytoplasm

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Q36. NEET 2021

Match List-I with List-II.

Open solution

	List-I		List-II
(a)	Cristae	(i)	Primary constriction in chromosome
(b)	Thylakoids	(ii)	Disc-shaped sacs in Golgi apparatus
(c)	Centromere	(iii)	Infoldings in mitochondria
(d)	Cisternae	(iv)	Flattened membranous sacs in stroma of plastids

Choose the correct answer from the options given below.

(A)

(a)	(b)	(c)	(d)
(ii)	(iii)	(iv)	(i)

(B)

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

(C)

(a)	(b)	(c)	(d)
(i)	(iv)	(iii)	(ii)

(D)

(a)	(b)	(c)	(d)
(iii)	(iv)	(i)	(ii)

Q37. NEET 2021

When the centromere is situated in the middle of two equal arms of chromosomes, the chromosome is referred as :

Open solution

(A) Acrocentric (B) Metacentric (C) Telocentric (D) Sub-metacentric

Q38. NEET 2021

The organelles that are included in the endomembrane system are

Open solution

(A) Golgi complex, Endoplasmic reticulum, Mitochondria and Lysosomes

(B) Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes

(C) Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

(D) Golgi complex, Mitochondria, Ribosomes and Lysosomes

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Q39. NEET 2020 Phase 2

Inclusion bodies of blue-green, purple and green photosynthetic bacteria are :

(A) Microtubules
(B) Contractile vacuoles
(C) Gas vacuoles
(D) Centrioles

Open solution

Q40. NEET 2020 Phase 2

The biosynthesis of ribosomal RNA occurs in :

(A) Nucleolus
(B) Ribosomes
(C) Golgi apparatus
(D) Microbodies

Open solution

Q41. NEET 2020 Phase 2

The size of Pleuropneumonia - like Organism (PPLO) is :

(A) $0.1\ \mu m$
(B) $0.02\ \mu m$
(C) $1-2\ \mu m$
(D) $10-20\ \mu m$

Open solution

Q42. NEET 2020 Phase 2

Match the following columns and select the correct option :

Column - I	Column - II
(a) Smooth Endoplasmic Reticulum	(i) Protein synthesis
(b) Rough endoplasmic reticulum	(ii) Lipid synthesis
(c) Golgi complex	(iii) Glycosylation
(d) Centriole	(iv) Spindle formation

(A) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
(C) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
(D) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Open solution

Q43. NEET 2020 Phase 1

Which of the following statements about inclusion bodies is incorrect?

(A) These are involved in ingestion of food particles
(B) They lie free in the cytoplasm
(C) These represent reserve material in cytoplasm
(D) They are not bound by any membrane

Open solution

Q44. NEET 2020 Phase 1

Which is the important site of formation of glycoproteins and glycolipids in eukaryotic cells?

(A) Peroxisomes
(B) Golgi bodies
(C) Polysomes
(D) Endoplasmic reticulum

Open solution

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Q45. NEET 2019 (Odisha)

Match the column I with column II.

Column I	Column II
(a) Golgi apparatus	(i) Synthesis of protein
(b) Lysosomes	(ii) Trap waste and excretory products
(c) Vacuoles	(iii) Formation of glycoproteins and glycolipids
(d) Ribosomes	(iv) Digesting biomolecules

Choose the right match from options given below :

(A) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)
(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
(C) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
(D) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

Open solution

Q46. NEET 2019 (Odisha)

Which of the following cell organelles is present in the highest number in secretory cells?

(A) Lysosome
(B) Mitochondria
(C) Golgi complex
(D) Endoplasmic reticulum

Open solution

Q47. NEET 2019 (Odisha)

Non-membranous nucleoplasmic structures in nucleus are the site for active synthesis of

(A) tRNA
(B) protein synthesis
(C) mRNA
(D) rRNA

Open solution

Q48. NEET 2019 (Odisha)

Which of the following nucleic acids is present in an organism having 70 S ribosomes only?

(A) Double stranded circular DNA with histone proteins
(B) Single stranded DNA with protein coat
(C) Double stranded circular naked DNA
(D) Double stranded DNA enclosed in nuclear membrane

Open solution

Q49. NEET 2019

Which of the following pair of organelles does not contain DNA?

(A) Lysosomes and Vacuoles
(B) Nuclear envelope and Mitochondria
(C) Mitochondria and Lysosomes
(D) Chloroplast and Vacuoles

Open solution

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Q50. NEET 2019

Which of the following statements regarding mitochondria is **incorrect**?

(A) Inner membrane is convoluted with infoldings

(B) Mitochondrial matrix contains single circular DNA molecule and ribosomes

(C) Outer membrane is permeable to monomers of carbohydrates, fats and proteins

(D) Enzymes of electron transport are embedded in outer membrane

Open solution

Q51. NEET 2019

Which of the following statements is **not** correct?

(A) Lysosomes are membrane bound structures.

(B) Lysosomes are formed by the process of packaging in the endoplasmic reticulum.

(C) Lysosomes have numerous hydrolytic enzymes.

(D) The hydrolytic enzymes of lysosomes are active under acidic pH.

Open solution

Q52. NEET 2019

The concept of "*Omnis cellula-e cellula*" regarding cell division was first proposed by :

(A) Schleiden

(B) Aristotle

(C) Rudolf Virchow

(D) Theodore Schwann

Open solution

Q53. NEET 2018

Which of the following is true for nucleolus?

(A) It takes part in spindle formation

(B) Larger nucleoli are present in dividing cells.

(C) It is a membrane-bound structure.

(D) It is a site for active ribosomal RNA synthesis.

Open solution

Q54. NEET 2018

The Golgi complex participates in

(A) activation of amino acid

(B) fatty acid breakdown

(C) formation of secretory vesicles

(D) respiration in bacteria

Open solution

Q55. NEET 2018

Which of the following events does not occur in rough endoplasmic reticulum?

(A) Cleavage of signal peptide

(B) Phospholipid synthesis

(C) Protein folding

(D) Protein glycosylation

Open solution

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Q56. NEET 2018

Select the incorrect match.

(A) Allosomes – Sex chromosomes

(B) Sub-metacentric chromosomes – L-shaped chromosomes

(C) Lampbrush chromosomes – Diplotene bivalents

(D) Polytene chromosomes – Oocytes of amphibians

Open solution

Q57. NEET 2017

Which of the following cell organelles is responsible for extracting energy from carbohydrates to form ATP?

(A) Ribosome (B) Mitochondrion (C) Chloroplast (D) Lysosome

Open solution

Q58. NEET 2016 Phase 2

Select the mismatch.

(A) Large central vacuoles – Animal cells (B) Methanogens –Prokaryotes

(C) Gas vacuoles – Green bacteria (D) Protists – Eukaryotes

Open solution

Q59. NEET 2016 Phase 2

Select the wrong statement-

(A) Pili and fimbriae are mainly involved in motility of bacterial cells

(B) Mycoplasma is a wall-less microorganism

(C) Bacterial cell wall is made up of peptidoglycan

(D) Cyanobacteria lack flagellated cells.

Open solution

Q60. NEET 2016 Phase 2

A cell organelle containing hydrolytic enzymes is

(A) Ribosome (B) Mesosome (C) Lysosome (D) Microsome

Open solution

Q61. NEET 2016 Phase 1

Mitochondria and chloroplast are-

(a) semi-autonomous organelles

(b) formed by division of pre existing organelles and they contain DNA but lack protein synthesising machinery

Which one of the following options is correct?

(A) Both (a) and (b) are false (B) Both (a) and (b) are correct

(C) (b) is true but (a) is false (D) (a) is true but (b) is false

Open solution

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Q62. NEET 2016 Phase 1

Microtubules are the constituents of

(A) Cilia, Flagella and Peroxisomes
(B) Centrosome, Nucleosome and Centrioles
(C) Spindle fibres, Centrioles and Cilia
(D) Centrioles, spindle fibres and chromatin

Open solution

Q63. NEET 2016 Phase 1

Which one of the following cell organelles is enclosed by a single membrane?

(A) Mitochondria (B) Chloroplasts (C) Lysosomes (D) Nuclei

Open solution

Q64. AIPMT 2015

Match the columns and identify the correct option.

Column I	Column II
(a) Thylakoids	(i) Disc-shaped sacs in Golgi apparatus
(b) Cristae	(ii) Condensed structure of DNA
(c) Cisternae	(iii) Flat membranous sacs in stroma
(d) Chromatin	(iv) Infoldings in mitochondria

(A) (a) → (iv), (b) → (ii), (c) → (i), (d) → (iii)
(B) (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii)
(C) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)
(D) (a) → (iii), (b) → (i), (c) → (iv), (d) → (ii)

Open solution

Q65. AIPMT 2015

Which of the following structures is not found in a prokaryotic cell?

(A) Mesosome
(B) Nuclear envelope
(C) Ribosome
(D) Plasma membrane

Open solution

Q66. AIPMT 2015

Cellular organelles with membranes are -

(A) endoplasmic reticulum, ribosomes and nuclei
(B) lysosomes, Golgi apparatus and mitochondria
(C) nuclei, ribosomes and mitochondria
(D) chromosomes, ribosomes and endoplasmic reticulum

Open solution

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Q67. AIPMT 2015

Which of the following are not membrane-bound ?

(A) Ribosomes (B) Mesosomes (C) Lysosomes (D) Vacuoles

Open solution

Q68. AIPMT 2015 Cancelled Paper

DNA is not present in :

(A) Nucleus (B) Chloroplast (C) Mitochondria (D) Ribosomes

Open solution

Q69. AIPMT 2015 Cancelled Paper

Nuclear envelope is a derivative of :

(A) Membrane of Golgi complex (B) Smooth endoplasmic reticulum
(C) Rough endoplasmic reticulum (D) Microtubules

Open solution

Q70. AIPMT 2015 Cancelled Paper

The structures that are formed by stacking of organized flattened membranous sacs in the chloroplasts are

(A) Stroma lamellae (B) Cristae (C) Stroma (D) Grana

Open solution

Q71. AIPMT 2015 Cancelled Paper

Select the correct matching in the following pairs :

(A) Rough ER – Oxidation of fatty acids (B) Smooth ER – Oxidation of phospholipids
(C) Rough ER – Synthesis of glycogen (D) Smooth ER – Synthesis of lipids

Open solution

Q72. AIPMT 2015 Cancelled Paper

The chromosomes in which centromere is situated close to one end are :

(A) Telocentric (B) Metacentric (C) Sub-metacentric (D) Acrocentric

Open solution

Q73. AIPMT 2015 Cancelled Paper

Which one of the following is not an inclusion body found in prokaryotes?

(A) Cyanophycean granule (B) Phosphate granule
(C) Polysome (D) Glycogen granule

Open solution

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Q74. AIPMT 2014

The solid linear cytoskeletal elements having a diameter of 6 nm and made up of a single type of monomer are known as:

(A) Microfilaments
(B) Lamins
(C) Microtubules
(D) Intermediate filaments

Open solution

Q75. AIPMT 2014

The osmotic expansion of a cell kept in water is chiefly regulated by:

(A) Vacuoles
(B) Plastids
(C) Mitochondria
(D) Ribosomes

Open solution

Q76. AIPMT 2014

Match the following and select the **correct** answer :

(a) Centriole	(i) Infoldings in mitochondria
(b) Chlorophyll	(ii) Thylakoids
(c) Cristae	(iii) Nucleic acid
(d) Ribozymes	(iv) Basal body cilia or flagella

(A) (a) → (iv), (b) → (iii), (c) → (i), (d) → (ii)
(B) (a) → (i), (b) → (iii), (c) → (ii), (d) → (iv)
(C) (a) → (iv), (b) → (ii), (c) → (i), (d) → (iii)
(D) (a) → (i), (b) → (ii), (c) → (iv), (d) → (iii)

Open solution

Q77. NEET 2013 (Karnataka)

The term 'glycocalyx' is used for :

(A) bacterial cell glyco-engineered to possess N-glycosylated proteins
(B) a layer present between cell wall and membrane of bacteria
(C) a layer surrounding the cell wall of bacteria
(D) cell wall of bacteria

Open solution

Q78. NEET 2013 (Karnataka)

Which of the following types of plastids does not contain stored food material ?

(A) Elaioplasts
(B) Chromoplasts
(C) Amyloplasts
(D) Aleuroplasts

Open solution

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Q79. NEET 2013 (Karnataka)

Which of the following best illustrates "FEEDBACK" in development ?

(A) As tissue (X) develops, it secretes something that induces tissue (Y) to develop

(B) As tissue (X) develops, it secretes something that slows down the growth of tissue (Y)

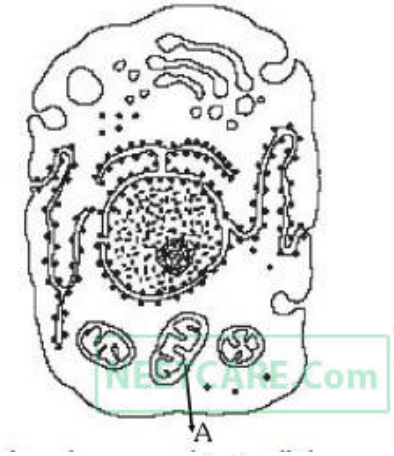
(C) Tissue (X) secretes RNA which changes the development of tissue (Y)

(D) As tissue (X) develops, it secretes enzymes that inhibit the development of tissue (Y)

Open solution

Q80. NEET 2013 (Karnataka)

Select the alternative giving correct identification and function of the organelle 'A' in the diagram



(A) Golgi body - provides packaging material

(B) Mitochondria - produce cellular energy in the form of ATP

(C) Endoplasmic reticulum - synthesis of lipids

(D) Lysosomes - secrete hydrolytic enzymes

Open solution

Q81. NEET 2013

The Golgi complex plays a major role :

(A) as energy transferring organelles

(B) in post translational modification of proteins and glycosidation of lipids

(C) in trapping the light and transforming it into chemical energy

(D) in digesting proteins and carbohydrates

Open solution

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Q82. NEET 2013

Which one of the following organelle in the figure correctly matches with its function ?



Open solution

(A) Golgi apparatus, formation of glycolipids
(B) Rough endoplasmic reticulum, formation of glycoproteins
(C) Rough endoplasmic reticulum, protein synthesis
(D) Golgi apparatus, protein synthesis

Q83. NEET 2013

A major site for synthesis of lipids is :

(A) RER (B) Nucleoplasm (C) SER (D) Symplast

Open solution

Q84. AIPMT 2012 Mains

Which one of the following cellular parts is correctly described ?

(A) Lysosomes - optimally active at a pH of about 8.5
(B) Centrioles - sites for active RNA synthesis
(C) Thylakoids - flattened membranous sacs forming the grana of chloroplasts
(D) Ribosomes - those on chloroplasts are larger (80s) while those in the cytoplasm are smaller (70s)

Open solution

Q85. AIPMT 2012 Mains

Which one of the following structures is an organelle within an organelle ?

(A) Mesosome (B) Peroxisome (C) Ribosome (D) ER

Open solution

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Q86. AIPMT 2012 Prelims

Select the the correct statement from the following regarding cell membrane

(A) Na^+ and K^+ ions move across cell membrane by passive transport

(B) Lipids are arranged in bilayer with polar heads towards the inner part

(C) Proteins make up 60 to 70% of the cell membrane

(D) Fluid mosaic model of cell membrane was proposed by singer and Nicolson

Open solution

Q87. AIPMT 2012 Prelims

What is true about ribosomes ?

(A) These are self - splicing introns of some RNAs.

(B) These are found only in eukaryotic cells

(C) These are composed of ribonucleic acid and proteins

(D) The prokaryotic ribosomes are 80 S, where 'S' stands for sedimentation coefficient.

Open solution

Q88. AIPMT 2012 Prelims

Which one of the following does not differ in E.coli and Chlamydomonas ?

(A) Chromosomal Organization (B) Cell wall

(C) Ribosomes (D) Cell membrane

Open solution

Q89. AIPMT 2011 Mains

Which one of the following is not considered as a part of the endomembrane system ?

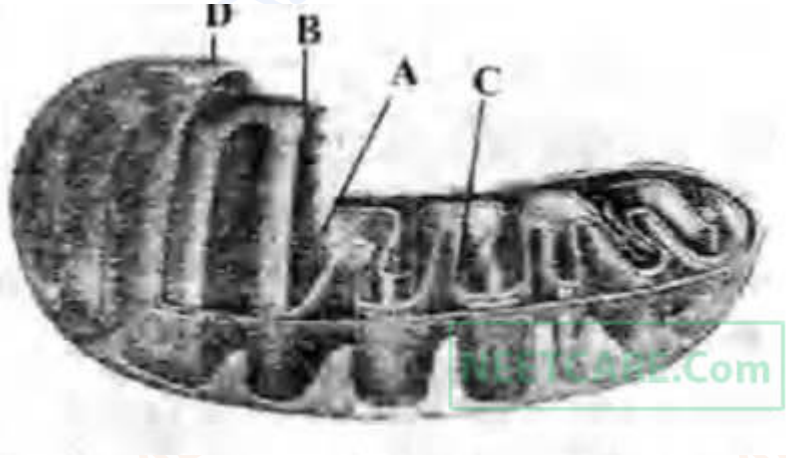
(A) Vacuole (B) Lysosome (C) Golgi complex (D) Peroxisome

Open solution

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Q90. AIPMT 2011 Mains

The figure below shows the structure of a mitochondrion with its four parts labelled (A), (B), (C) and (D) Select the part correctly matched with its function. –



Open solution

(A) Part (B) : Inner membrane – forms infoldings called cristae

(B) Part (C) : Cristae – possess single circular DNA molecule and ribosome

(C) Part (A) : Matrix – major site for respiratory chain enzymes

(D) Part (D) : Outer membrane – gives rise to inner membrane by splitting

Q91. AIPMT 2011 Prelims

Peptide synthesis inside a cell takes place in :

(A) Chromoplast (B) Ribosomes (C) Chloroplast (D) Mitochondria

Open solution

Q92. AIPMT 2011 Prelims

Important site for formation of glycoproteins and glycolipids is :

(A) Plastid (B) Lysosome (C) Golgi apparatus (D) Vacuole

Open solution

Q93. AIPMT 2010 Mains

An elaborate network of filamentous proteinaceous structures present in the cytoplasm which helps in the maintenance of cell shape is called -

(A) Cytoskeleton (B) Endosplasmic Reticulum

(C) Thylakoid (D) Plasmalemma

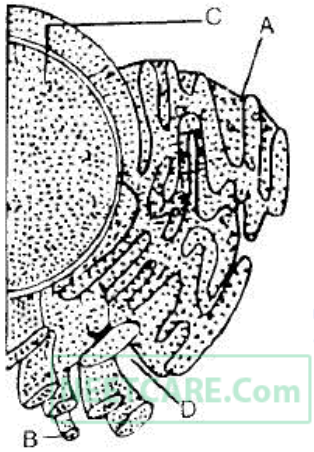
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Q94. AIPMT 2010 Mains

Identify the components labelled A, B, C and D in the diagram below from the list (i) to (viii) given along with –

Open solution



Components :
(i) Cristae of mitochondria
(ii) Inner membrane of mitochondria
(iii) Cytoplasm
(iv) Smooth endoplasmic reticulum
(v) Rough endoplasmic reticulum
(vi) Mitochondrial matrix
(vii) Cell vacuole
(viii) Nucleus
The correct components are :

(A) (A) → (v), (B) → (iv), (C) → (viii), (D) → (iii)

(B) (A) → (i), (B) → (iv), (C) → (viii), (D) → (vi)

(C) (A) → (v), (B) → (i), (C) → (iii), (D) → (ii)

(D) (A) → (vi), (B) → (v), (C) → (iv), (D) → (vii)

Q95. AIPMT 2010 Prelims

The plasma membrane consists mainly of –

Open solution

(A) proteins embedded in a carbohydrate bilayer

(B) phospholipids embedded in a protein bilayer

(C) proteins embedded in a phospholipid bilayer

(D) proteins embedded in a polymer of glucose molecules

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Q96. AIPMT 2010 Prelims

The main area of various types of activities of a cell is –

(A) Cytoplasm
(B) Plasma membrane
(C) Nucleus
(D) Mitochondrion

Open solution

Q97. AIPMT 2010 Prelims

Which one of the following has its own DNA?

(A) Dictyosome
(B) Mitochondria
(C) Peroxisome
(D) Lysosome

Open solution

Q98. AIPMT 2010 Prelims

Which one of the following structures between two adjacent cells is an effective transport pathway?

(A) Endoplasmic reticulum
(B) Plasmodesmata
(C) Plasmalemma
(D) Plastoquinones

Open solution

Q99. AIPMT 2009

Middle lamella is composed mainly of :

(A) Calcium pectate
(B) Phosphoglycerides
(C) Muramic acid
(D) Hemicellulose

Open solution

Q100. AIPMT 2009

Cytoskeleton is made up of :

(A) Cellulose microfibrils
(B) Proteinaceous filaments
(C) Calcium carbonate granules
(D) Callose deposits

Open solution

Q101. AIPMT 2009

Plasmodesmata are :

(A) Locomotory structures
(B) Connection between adjacent cells
(C) Membranes connecting the nucleus with plasmalemma
(D) Lignified cemented layers between cells

Open solution

Q102. AIPMT 2008

In germinating seeds fatty acids are degraded exclusively in the:

(A) proplastids
(B) peroxisomes
(C) glyoxysomes
(D) mitochondria

Open solution

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Q103. AIPMT 2008

Vacuole in a plant cell:

(A) is membrane-bound and contains storage proteins and lipids

(B) is membrane-bound and contains water and excretory substances

(C) lacks membrane and contains air

(D) Lacks membrane and contains water and excretory substances

Open solution

Q104. AIPMT 2008

The two sub-units of ribosome remain united at a critical ion level of :

(A) magnesium (B) manganese (C) calcium (D) copper

Open solution

Q105. AIPMT 2008

Keeping in view the "fluid mosaic model" for the structure of cell membrane, which one of the following statements is correct with respect to the movement of lipids and proteins from one lipid monolayer to the other (described as flipflop movement)?

(A) Both lipids and proteins can flip-flop

(B) While lipids can rarely flip-flop proteins cannot

(C) While proteins can flip-flop, lipids cannot

(D) neither lipids, nor proteins can flip-flop

Open solution

Q106. AIPMT 2007

Which one of the following is not a constituent of cell membrane ?

(A) Glycolipids (B) Proline (C) Phospholipids (D) Cholesterol

Open solution

Q107. AIPMT 2007

Select the wrong statement from the following:

(A) The chloroplasts are generally much larger than mitochondria

(B) Both chloroplasts and mitochondria contain DNA

(C) Both chloroplasts and mitochondria contain an inner and an outer membrane

(D) Both chloroplasts and mitochondria have an internal compartment, the thylakoid space bounded by the thylakoid membrane

Open solution

Q108. AIPMT 2007

Biological organization starts with:-

(A) Cellular level (B) Atomic level

(C) Organismic level (D) Submicroscopic molecular level

Open solution

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Q109. AIPMT 2006

Which of the following statements regarding mitochondrial membrane is not correct ?

(A) The inner membrane is highly convoluted forming a series of infoldings

(B) The outer membrane resembles a sieve

(C) The outer membrane is permeable to all kinds of molecules

(D) The enzymes of the electron transfer chain are embedded in the outer membrane

Open solution

Q110. AIPMT 2006

Which of the following statements regarding cilia is not correct ?

(A) Microtubules of cilia are composed of tubulin

(B) The organized beating of cilia is controlled by fluxes of Ca^{2+} across the membrane

(C) Cilia are hair-like cellular appendages

(D) Cilia contain an outer ring of nine doublet microtubules surrounding two single microtubules

Open solution

Q111. AIPMT 2006

A major breakthrough in the studies of cells came with the development of electron microscope. This is because

(A) Electron beam can pass through thick materials, whereas light microscopy requires thin sections

(B) The resolution power of the electron microscope is much higher than that of the light microscope

(C) The electron microscope is more powerful than the light microscope as it uses a beam of electrons which has wavelength much longer than that of photons

(D) The resolving power of the electron microscope is 200-350 nm as compared to 0.1-0.2 nm for the light microscope

Open solution

Q112. AIPMT 2005

Chlorophyll in chloroplasts is located in

(A) Stroma (B) Grana (C) Pyrenoid (D) Both (A) and (C)

Open solution

Q113. AIPMT 2005

A student wishes to study the cell structure under a light microscope having 10X eyepiece and 45X objective. He should illuminate the object by which one of the following colours of light so as to get the best possible resolution ?

(A) Yellow (B) Red (C) Blue (D) Green

Open solution

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Q114. AIPMT 2005

According to widely accepted "fluid mosaic model" cell membranes are semi-fluid, where lipids and integral proteins can diffuse randomly. In recent years, this model has been modified in several respects. In this regard, which of the following statements are incorrect ?

(A) Proteins in cell membranes can travel within the lipid bilayer

(B) Proteins can also undergo flip-flop movements in the lipid bilayer

(C) Proteins can remain confined within certain domains of the membranes

(D) Many proteins remain completely embedded within the lipid bilayer

Open solution

Q115. AIPMT 2005

Centromere is required for -

(A) Crossing over

(B) Movement of chromosomes towards poles

(C) Transcription

(D) Cytoplasmic cleavage

Open solution

Q116. AIPMT 2005

Chemiosmotic theory of ATP synthesis in the chloroplasts and mitochondria is based on

(A) Proton gradient

(B) Membrane potential

(C) Accumulation of Na^+ ions

(D) Accumulation of K^+ ions

Open solution

Q117. AIPMT 2005

The main organelle involved in modification and routing of newly synthesized proteins to their destinations is -

(A) Mitochondria

(B) Endoplasmic Reticulum

(C) Chloroplast

(D) Lysosome

Open solution

Q118. AIPMT 2004

The telomeres of eukaryotic chromosomes consist of short sequences of

(A) Adenine rich repeats

(B) Cytosine rich repeats

(C) Thymine rich repeats

(D) Guanine rich repeats

Open solution

Q119. AIPMT 2004

In chloroplasts, chlorophyll is present in the :-

(A) Outer membrane (B) Inner membrane (C) Thylakoids (D) Stroma

Open solution

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Q120. AIPMT 2004

Flagella of prokaryotic and eukaryotic cells differ in

(A) type of movement and placement in cell

(B) location in cell and mode of functioning

(C) micro-tubular organisation and type of movement

(D) micro-tubular organisation and function

Open solution

Q121. AIPMT 2004

Extranuclear inheritance is a consequence of presence of genes in

(A) mitochondria and chloroplasts

(B) endoplasmic reticulum and mitochondria

(C) ribosomes and chloroplast

(D) lysosomes and ribosomes

Open solution

Q122. AIPMT 2003

Cellular totipotency is demonstrated by :-

(A) All eukaryotic cells

(B) Only gymnosperm cells

(C) Only bacterial cells

(D) All plant cells

Open solution

Q123. AIPMT 2002

Ribosomes are produced in :

(A) Mitochondria

(B) Golgibody

(C) Nucleolus

(D) Cytoplasm

Open solution

Q124. AIPMT 2002

In fluid mosaic model of plasma membrane

(A) Phospholipids form a bimolecular layer in middle part

(B) Upper layer is non-polar and hydrophilic

(C) Proteins form a middle layer

(D) Polar layer is hydrophobic

Open solution

Q125. AIPMT 2002

Which of the following occurs more than one and less than five in a chromosome?

(A) Chromatid

(B) Chromomere

(C) Centromere

(D) Telomere

Open solution

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Q126. AIPMT 2001

Element necessary for the middle lamella is

(A) K (B) Cu (C) Ca (D) Zn

Open solution

Q127. AIPMT 2001

Microtubules are absent in : -

(A) Spindle fibres (B) Mitochondria (C) Centriole (D) Flagella

Open solution

Q128. AIPMT 2001

Proteinaceous pigment which controls the activities concerned with light is

(A) Carotenoids (B) Phytochrome (C) Chlorophyll (D) Anthocyanin

Open solution

Q129. AIPMT 2001

Extranuclear chromosomes occur in

(A) peroxisome, ribosome
(B) chloroplast and mitochondria
(C) mitochondria and ribosome
(D) chloroplast and lysosome

Open solution

Q130. AIPMT 2000

Lysosome contains :

(A) Reductive enzymes
(B) Oxidative enzymes
(C) Anabolic enzymes
(D) Hydrolytic enzymes

Open solution

Q131. AIPMT 2000

Which of the following ribosomes are engaged in protein synthesis in animal cell ?

(A) Ribosomes of only nucleolus and cytosol
(B) Ribosomes which occur on nuclear membrane and E.R.
(C) Ribosomes of only mitochondria and cytosol
(D) Ribosomes of only cytosol

Open solution

Q132. AIPMT 2000

Function of telomeres in nucleus is

(A) To seal the ends of chromosome
(B) Poleward movement
(C) To recognize the homologous chromosome
(D) To initiate the RNA synthesis

Open solution

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Q133. AIPMT 2000

Which cell organelle is concerned with glycosylation of protein ?

(A) Endoplasmic reticulum
(B) Ribosome
(C) Mitochondria
(D) Peroxisome

Open solution

Q134. AIPMT 2000

Extranuclear DNA (genes) are located in

(A) lysosomes and chloroplasts
(B) Golgi complex and ribosomes
(C) chloroplasts and mitochondria
(D) ribosomes and mitochondria

Open solution

Q135. AIPMT 2000

'Signal hypothesis' for the biosynthesis of secretory type of proteins was proposed by

(A) Camillo Golgi
(B) Blobel and Sabatini
(C) Baltimore
(D) Sheeler and Bianchi

Open solution

Q136. AIPMT 1999

Photosynthetic bacteria have pigments in :

(A) chromoplasts
(B) leucoplasts
(C) chloroplasts
(D) chromatophore

Open solution

Q137. AIPMT 1999

In prokaryotes, the genetic material is

(A) linear DNA without histones
(B) linear DNA with histones
(C) circular DNA with histones
(D) circular DNA without histones

Open solution

Q138. AIPMT 1999

The proteins are synthesized at :

(A) ribosomes
(B) mitochondria
(C) centrosomes
(D) Golgi bodies

Open solution

Q139. AIPMT 1999

Which of the following has single membrane ?

(A) Mitochondria
(B) Nucleus
(C) Cell wall
(D) Spherosomes

Open solution

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Q140. AIPMT 1999

Some of the enzymes, which are associated in converting fats into carbohydrates are present in :

(A) microsomes (B) liposomes (C) Golgi bodies (D) glyoxysomes

Open solution

Q141. AIPMT 1998

The DNA of *E. coli* is

(A) single stranded and linear (B) single stranded and circular

(C) double stranded and linear (D) double stranded and circular

Open solution

Q142. AIPMT 1998

Microtubule is involved in the

(A) cell division (B) DNA recognition

(C) muscle contraction (D) membrane architecture

Open solution

Q143. AIPMT 1998

How many genome types are present in a typical green plant's cell?

(A) Two (B) Three (C) More than five (D) More than ten

Open solution

Q144. AIPMT 1997

The site of respiration in bacteria is

(A) ribosome (B) microsome (C) episome (D) mesosome

Open solution

Q145. AIPMT 1997

The hereditary material present in the bacterium *E.coli* is

(A) single-stranded DNA (B) double-stranded DNA

(C) DNA (D) RNA

Open solution

Q146. AIPMT 1997

Which of the following structures will not be common to mitotic cell of a higher plant?

(A) Centriole (B) Spindle fibre (C) Cell plate (D) Centromere

Open solution

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Q147. AIPMT 1997

Protein synthesis in an animal cell, takes place

(A) in the cytoplasm as well as endoplasmic reticulum

(B) only on ribose attached to nucleon

(C) only in the cytoplasm

(D) in the nucleolus as well as in the cytoplasm

Open solution

Q148. AIPMT 1996

What are the sex organs provided in some bacteria?

(A) Sex pili (B) Plasmid (C) Circular DNA (D) Gametes

Open solution

Q149. AIPMT 1996

Which type of DNA is found in bacteria?

(A) Circular free DNA (B) Membrane bound DNA

(C) Straight DNA (D) Helical DNA

Open solution

Q150. AIPMT 1996

Lysosomes are rich in

(A) nucleic acids (B) hydrolytic enzymes

(C) carbohydrates (D) hormones

Open solution

Q151. AIPMT 1995

The prokaryotic flagella possess

(A) helically arranged protein molecule (B) “9 + 2” membrane enclosed structure

(C) unit membrane enclosed fibre (D) protein membrane enclosed fibre

Open solution

Q152. AIPMT 1994

Poisons like cyanide inhibit Na^+ efflux and K^+ influx during cellular transport. This inhibitory effect is reversed by an injection of ATP. This demonstrates that

(A) ATP is the carrier protein in the transport system

(B) energy for Na^+-K^+ exchange pump comes from ATP

(C) ATP is hydrolysed by ATPase to release energy

(D) Na^+-K^+ exchange pump operates in the cell

Open solution

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Q153. AIPMT 1994

Which of the following organelles contain enzymes that have digestive action?

(A) Ribosomes (B) Polysomes (C) Plastids (D) Lysosomes

Open solution

Q154. AIPMT 1994

In mitochondria, cristae act as sites for

(A) protein synthesis
(B) phosphorylation of flavoproteins
(C) breakdown of macromolecules
(D) oxidation-reduction reaction

Open solution

Q155. AIPMT 1994

Centromere is required for

(A) replication of DNA
(B) chromosome segregation
(C) poleward movement of chromosomes
(D) cytoplasmic cleavage

Open solution

Q156. AIPMT 1994

Which one of the following organelles is located near the nucleus and contains a collection of flattened membrane bound cisternae?

(A) Nucleolus (B) Mitochondrion (C) Centriole (D) Golgi apparatus

Open solution

Q157. AIPMT 1994

The inner membrane of the mitochondria is, usually, highly convoluted forming a series of infoldings known as

(A) thylakoids (B) lamellae (C) cristae (D) grana

Open solution

Q158. AIPMT 1994

Besides giving out secretory vesicles, the Golgi apparatus is also concerned with the formation of

(A) lysosomes
(B) plastids
(C) grana of chloroplasts
(D) cell plates after cell division in plants

Open solution

Q159. AIPMT 1993

Genophore/bacterial genome or nucleoid is made of

(A) histones and nonhistones
(B) RNA and histones
(C) a single double stranded DNA
(D) a single stranded DNA

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Q160. AIPMT 1993

Active and passive transports across cell membrane differ in

(A) passive transport is nonselective

(B) passive transport is along the concentration gradient while active transport is due to metabolic energy

(C) active transport is more rapid

(D) passive transport is confined to anions while active transport in confined to cations

Open solution

Q161. AIPMT 1993

Balbiani rings (puffs) are sites of

(A) DNA replication

(B) RNA and protein synthesis

(C) synthesis of polysaccharides

(D) synthesis of lipids

Open solution

Q162. AIPMT 1993

In plant cells, peroxisomes are associated with

(A) photorespiration (B) phototropism (C) photoperiodism (D) photosynthesis

Open solution

Q163. AIPMT 1993

Membranous bag with hydrolytic enzymes which is used for controlling intracellular digestion of macromolecules is

(A) endoplasmic reticulum

(B) nucleosome

(C) lysosome

(D) phagosome

Open solution

Q164. AIPMT 1993

Golgi apparatus is absent in

(A) higher plants

(B) yeast

(C) bacteria and blue-green algae

(D) none

Open solution

Q165. AIPMT 1993

Cell recognition and adhesion occur due to biochemicals of cell membranes named

(A) proteins

(B) lipids

(C) proteins and lipids

(D) glycoproteins and glycolipids

Open solution

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Q166. AIPMT 1993

In salivary gland chromosomes/polytene chromosomes, pairing is

(A) absent
(B) occasional
(C) formed between non-homologous chromosomes
(D) formed between homologous chromosomes

Open solution

Q167. AIPMT 1993

Which is correct about cell theory in view of current status of our knowledge about cell structure?

(A) It needs modification due to discovery of subcellular structures like chloroplasts and mitochondria
(B) Modified cell theory means that all living being are composed of cells capable of reproducing
(C) Cell theory does not hold good because all living beings (*e.g.*, viruses) do not have cellular organisation
(D) Cell theory means that all living objects consists of cells whether or not capable of reproducing

Open solution

Q168. AIPMT 1993

Names of Schleiden and Schwann are associated with

(A) protoplasm as the physical basis of life
(B) cell theory
(C) theory of cell lineage
(D) nucleus functions as control centre of cell

Open solution

Q169. AIPMT 1992

All types of plastids possess essentially the same structure because they

(A) perform the same function
(B) store food materials like starch, fat and protein
(C) occur in aerial parts
(D) can transform from one form to another

Open solution

Q170. AIPMT 1992

Experiments on *Acetabularia* by Hammerling proved the role of

(A) cytoplasm in controlling differentiation
(B) nucleus in heredity
(C) chromosomes in heredity
(D) nucleo-cytoplasmic ratio

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Q171. AIPMT 1992

Which one is *apparato reticolare*?

(A) Golgi apparatus
(B) Endoplasmic reticulum
(C) Microfilaments
(D) Microtubules

Open solution

Q172. AIPMT 1992

An outer covering membrane is absent over

(A) nucleolus
(B) lysosome
(C) mitochondrion
(D) plastid

Open solution

Q173. AIPMT 1992

All plastids have similar structure because they can

(A) store starch, lipids and proteins
(B) get transformed from one type to another
(C) perform same function
(D) be present together

Open solution

Q174. AIPMT 1992

Oxysomes or $F_0 - F_1$ particles occur on

(A) thylakoids
(B) mitochondrial surface
(C) inner mitochondrial membrane
(D) chloroplast surface

Open solution

Q175. AIPMT 1992

Ribosomes are the centre for

(A) respiration
(B) photosynthesis
(C) protein synthesis
(D) fat synthesis

Open solution

Q176. AIPMT 1992

Angstrom (Å) is equal to

(A) $0.01\ \text{mm}$
(B) $0.001\ \text{mm}$
(C) $0.0001\ \text{mm}$
(D) $0.00001\ \text{mm}$

Open solution

Q177. AIPMT 1992

Electron microscope has a high resolution power. This is due to

(A) electromagnetic lenses
(B) very low wavelength of electron beam
(C) low wavelength of light source used
(D) high numerical aperture of glass lenses used

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Q178. AIPMT 1991

Addition of new cell wall particles amongst the existing ones is

(A) deposition (B) apposition (C) intussusception (D) aggregation

Open solution

Q179. AIPMT 1991

Cell wall shows

(A) complete permeability (B) semipermeability

(C) differential permeability (D) impermeability

Open solution

Q180. AIPMT 1991

Ribosomes were discovered by

(A) Golgi (B) Porter (C) de Robertis (D) Palade

Open solution

Q181. AIPMT 1991

Fluid mosaic model of cell membrane was put forward by

(A) Danielli and Davson (B) Singer and Nicolson

(C) Garner and Allard (D) Watson and Crick

Open solution

Q182. AIPMT 1991

Resolution power is the ability to

(A) distinguish two close points (B) distinguish two close objects

(C) distinguish amongst organelles (D) magnify image

Open solution

Q183. AIPMT 1990

The main difference in Gram (+)ve and Gram (−)ve bacteria resides in their

(A) cell wall (B) cell membrane (C) cytoplasm (D) flagella

Open solution

Q184. AIPMT 1990

Electron microscope has a high resolution power. This is due to

(A) electromagnetic lenses

(B) very low wavelength of electron beam

(C) low wavelength of light source used

(D) high numerical aperture of glass lenses used

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Q185. AIPMT 1990

Hammerling's experiments of *Acetabularia* involved exchanging

(A) cytoplasm
(B) nucleus
(C) rhizoid and stalk
(D) gametes

Open solution

Q186. AIPMT 1990

The latest model for plasma membrane is

(A) lamellar model
(B) unit membrane model
(C) fluid mosaic model
(D) molecular lipid model

Open solution

Q187. AIPMT 1990

Magnification of compound microscope is not connected with

(A) numerical aperture
(B) focal length of objective
(C) focal length of eye piece
(D) tube length

Open solution

Q188. AIPMT 1989

Nucleoproteins are synthesised in

(A) nucleoplasm (B) nuclear envelope (C) nucleolus (D) cytoplasm

Open solution

Q189. AIPMT 1989

Polyribosomes are aggregates of

(A) ribosomes and rRNA
(B) only rRNA
(C) peroxisomes
(D) several ribosomes held together by string of mRNA

Open solution

Q190. AIPMT 1989

Plasma membrane is made of

(A) proteins and carbohydrates
(B) proteins and lipids
(C) proteins, lipids and carbohydrates
(D) proteins, some nucleic acid and lipids

Open solution

Q191. AIPMT 1989

Organelles can be separated from cell homogenate through

(A) chromatography
(B) X-rays diffraction
(C) differential centrifugation
(D) auto-radiography

Open solution

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Q192. AIPMT 1988

Acetabularia used in Hammerling's nucleocytoplasmic experiments is

(A) unicellular fungus
(B) multicellular fungus
(C) unicellular uninucleate green algae
(D) unicellular multinucleate green algae

Open solution

Q193. AIPMT 1988

According to fluid mosaic model, plasma membrane is composed of

(A) phospholipids and oligosaccharides
(B) phospholipids and hemicellulose
(C) phospholipids and integral proteins
(D) phospholipids, extrinsic proteins and intrinsic proteins

Open solution

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Answer Key

1. **B**	2. **B**	3. **C**	4. **C**	5. **A**
6. **C**	7. **B**	8. **C**	9. **A**	10. **D**
11. **C**	12. **D**	13. **C**	14. **C**	15. **B**
16. **A**	17. **C**	18. **D**	19. **B**	20. **D**
21. **A**	22. **B**	23. **D**	24. **C**	25. **C**
26. **B**	27. **D**	28. **A**	29. **B**	30. **C**
31. **C**	32. **C**	33. **A**	34. **C**	35. **B**
36. **D**	37. **B**	38. **C**	39. **C**	40. **A**
41. **A**	42. **B**	43. **A**	44. **B**	45. **B**
46. **C**	47. **D**	48. **C**	49. **A**	50. **D**
51. **B**	52. **C**	53. **D**	54. **C**	55. **B**
56. **D**	57. **B**	58. **A**	59. **A**	60. **C**
61. **D**	62. **C**	63. **C**	64. **C**	65. **B**
66. **B**	67. **A**	68. **D**	69. **C**	70. **D**
71. **D**	72. **D**	73. **C**	74. **A**	75. **A**
76. **C**	77. **C**	78. **B**	79. **A**	80. **B**
81. **B**	82. **C**	83. **C**	84. **C**	85. **C**
86. **D**	87. **C**	88. **D**	89. **D**	90. **A**
91. **B**	92. **C**	93. **A**	94. **A**	95. **C**
96. **A**	97. **B**	98. **B**	99. **A**	100. **B**
101. **B**	102. **C**	103. **B**	104. **A**	105. **B**
106. **B**	107. **D**	108. **D**	109. **D**	110. **B**
111. **D**	112. **B**	113. **C**	114. **B**	115. **B**
116. **A**	117. **B**	118. **D**	119. **C**	120. **C**
121. **A**	122. **D**	123. **C**	124. **A**	125. **D**
126. **C**	127. **B**	128. **B**	129. **B**	130. **D**
131. **B**	132. **A**	133. **A**	134. **C**	135. **B**
136. **D**	137. **D**	138. **A**	139. **D**	140. **D**
141. **D**	142. **A**	143. **B**	144. **D**	145. **B**
146. **A**	147. **A**	148. **A**	149. **A**	150. **B**
151. **A**	152. **B**	153. **D**	154. **D**	155. **C**

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156. **D**	157. **C**	158. **A**	159. **C**	160. **B**
161. **B**	162. **A**	163. **C**	164. **C**	165. **D**
166. **D**	167. **C**	168. **B**	169. **D**	170. **B**
171. **A**	172. **A**	173. **B**	174. **C**	175. **C**
176. *Scan QR*	177. **B**	178. **C**	179. **A**	180. **D**
181. **B**	182. **B**	183. **A**	184. **B**	185. **C**
186. **C**	187. **A**	188. **D**	189. **D**	190. **C**
191. **C**	192. **C**	193. **D**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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